



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2014/2015

DIGITAL SCHOOL OF VIRTUAL AND OPEN LEARNING

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION

ECT 305: CHEMISTRY METHODS

DATE: Thursday 30th April 2015

TIME: 4.30p.m-6.30p.m

INSTRUCTIONS

Answer question 1 and any other TWO

Question 1

- a) i. Simple organic chemistry is taught at form three/form four. A student asks you:-

“Sir/Madam what product is formed between bromine water and the second member of the alkane series?”

What answer would you give? (4 marks)

- ii. The same student follow up with this question:-

“What chemical reason(s) determine the product(s) you have shown?”

What answer would you give? (4 marks)

- b) i. As a chemistry teacher you also double up as a laboratory manager in your school.

During one practical session a student pours acid on their skin. What first aid would you administer? (4 marks)

- ii. At another time a student has glass pieces lodged in their arm whilst trying to push tubing into a stopper. What action would you take to manage the situation? (4 marks)

- c) i. Evaluation in chemistry is a requirement of Kenyan secondary science. Briefly discuss TWO advantages and TWO disadvantages of evaluation. (4 marks)

- ii. Nitric acid reacts with sodium hydrogen carbonate. Write out the symbol equation of this reaction and a 4-step objective type question could accompany the equation in an exam. (5 marks)

- d) i. For a fire to occur in a laboratory certain conditions must be present. Which are these?
(5 marks)
- ii. What advice would you give to a technician who uses water to extinguish oil fires?
(5 marks)

Question 2

The following occurred whilst you were teaching the topic “Alkanes”. (S=Student)

S1. Please write for me the structural formulae for the first members of the alkane series.

(5 marks)

S2. What are their respective names.

(5 marks)

S3. From what you have written can we derive a general formula and what is it? (5 marks)

S4. It appears there is a recurring pattern between consecutive members of the family. Which is it?
(5 marks)

What answers would you give?

Question 3

- a) i. You are teaching temperature dependence upon chemical reactions. You choose to do so using a metal and acid.

Write out a balanced symbol equation of your select metal and acid. (5 marks)

- ii. You want to show how doubling the temperature speeds up the reaction. Briefly describe the experimental set-up for this investigation. (5 marks)

- b) i. You carry out this investigation but this time you want to show surface area dependence upon chemical reactions. Briefly discuss how you would set up the experiment. (5 marks)
- ii. Finally you want to teach the interface between temperature and surface area in chemical reactions. How would you set up the experiment?

Question 4

- a) i. As a chemistry teacher you also double up as your school’s laboratory manager.

How would you store the following;-

- Potassium metal
- Chlorine water
- Sodium water

- Sodium hydroxide solution (9 marks)
- ii. You want to buy reagents in small quantities. What three guide lines would you follow in doing so? (3 marks)
- b) i. You are teaching structure of chemical compounds as in ammonia and phosphine. You have a choice of using a chart or a ball and spoke model. Which would you use and why? (5 mark)
- ii. These two compounds have comparable chemical properties. What is the critical common determinant of their chemical reactions?

Question 5

- a) i. Chemistry is tied with social/environmental/political events that happen around us. We therefore must mention them to our learners. A student asks you;-
- “Sir/Madam The USA invaded Iraq in 2003 claiming that somehow they had dangerous chemical compounds. What is the common name of these compounds?”
- What answer would you give? (5 marks)
- ii. In the same note the student follows up by saying IAEA was involved in ratifying Iraq’s position then. What would you say the acronym IAEA stands for? (5 marks)
- b). Environmental degradation is now measured and all nations are expected to follow international standards. What would you tell your students the following measures are;-
- i. Carbon footprint (5 marks)
 - ii. Forest footprint (5 marks)